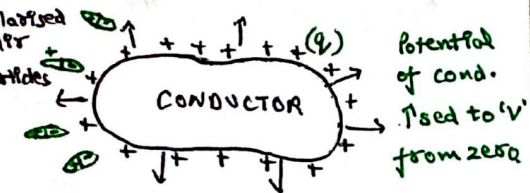


Capacitors - 1

Capacitance %

→ There is a capacity of every conductor upto which it can be charged.

→ If charged more than its capacity, then medium particles may get polarised and charge starts leaking from the conductor.



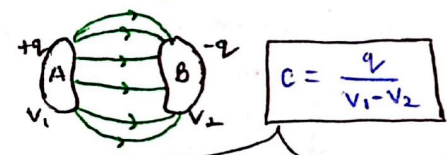
→ This charge limit is defined by the help of a physical constant known as Capacitance.

$$q = CV$$

Capacitance of conductor (Farad)

Capacitance of system of two conductors:

→ For a system of two conductors

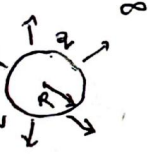


Do not depends on
→ charge
→ Potential diff

Depends on
→ Distance b/w plates
→ Shape/size of plates
→ Medium b/w them

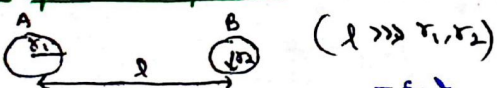
1) Conducting isolated sphere %

Here, the reference (zero volt point) is at infinity, so



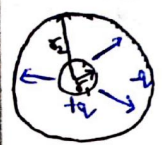
$$C = \frac{q}{V} = 4\pi\epsilon_0 R$$

2) Spherical capacitor %



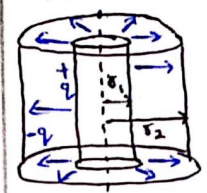
$$C = \frac{4\pi\epsilon_0 r_1 r_2}{r_1 + r_2}$$

3) Spherical shell capacitor %



$$C = \frac{4\pi\epsilon_0 r_1 r_2}{r_2 - r_1}$$

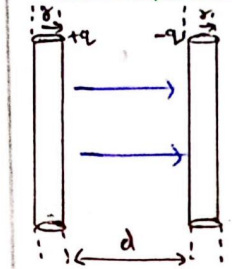
4) Cylindrical Capacitor %



$$C = \frac{2\pi\epsilon_0 l}{\ln(r_2/r_1)}$$

(l → length of cylinder)

5) Wire capacitor %

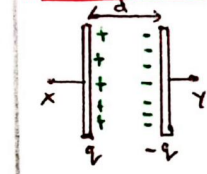


② charge density of 'long' wire is given.
 $\lambda = q/l$

$$C = \frac{\pi\epsilon_0 l}{\ln(d/r)}$$

(d >> r)

Parallel Plate Capacitor %



→ size of plates >> d

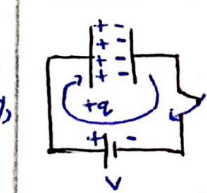
Here,

$$V = Ed = \left(\frac{\sigma}{\epsilon_0}\right) d$$

$$C = \frac{A\epsilon_0}{d}$$

$$V = \frac{q \cdot d}{A\epsilon_0}$$

Energy stored in a capacitor %



(i) Work done by battery,

$$W_B = (q)_{\text{flow}} \times V$$

$$W_B = CV^2$$

$$[\because (q)_{\text{flow}} = CV]$$

(ii) Energy stored b/w plates of capacitor, after complete charging

$$U = \frac{1}{2} CV^2 = \frac{1}{2} qV = \frac{q^2}{2C}$$

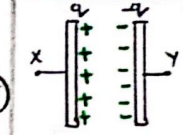
(iii) Heat dissipated = $W_B - U$

$$H = \frac{1}{2} CV^2 \quad [\text{Half energy is lost as heat}]$$

NOTE: Heat dissipation in charging a group of n-capacitors can directly be given by

$$H = \sum_{i=1}^n \frac{\Delta q_i^2}{2C_i} \quad (\Delta q \rightarrow \text{change in charge after charging})$$

Force b/w plate of a Capacitor %



Electric field due to any one plate of capacitor is,

$$|E_+| = |E_-| = \frac{q}{2A\epsilon_0}$$

→ So force of attraction on one plate due to other is given by,

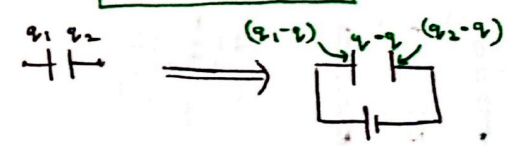
$$F = q|E| = \frac{q^2}{2A\epsilon_0}$$

Redistribution of charges %

→ Suppose the plates have q_1 & q_2 charge before charging.

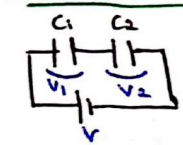
→ Then, after charging, the inner surfaces of plates will have equal & opp. 'q' charge given by

$$|q| = \frac{q_1 + q_2}{2} \quad [\text{Put } q_1 \text{ & } q_2 \text{ with sign}]$$



Combination of Capacitors %

1) Series Combination %



$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$$

2) Charge on all capacitors in series connection is same.

$$q = C_{eq} \cdot V$$

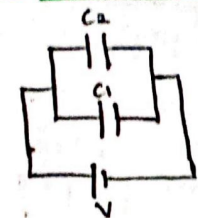
3) Potential diff. across each capacitor is in reverse ratio of their capacitance

$$\frac{V_1}{V_2} = \frac{C_2}{C_1} \quad V_1 = \frac{C_2 V}{C_1 + C_2} \quad V_2 = \frac{C_1 V}{C_1 + C_2}$$

1) If n -capacitors are connected in series, then

$$C_{eq} = \frac{C}{n} \quad \& \quad V_{each} = \frac{V}{n}$$

2) Parallel Combination :-



$$C_{eq} = C_1 + C_2 + \dots$$

2) Potential across all capacitors is same (V).

3) Charge on each capacitor is diff.

$$q_1 = C_1 V$$

$$q_2 = C_2 V$$

Charges divide in the ratio of their capacitance, i.e.

$$\frac{q_1}{q_2} = \frac{C_1}{C_2}$$

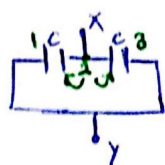
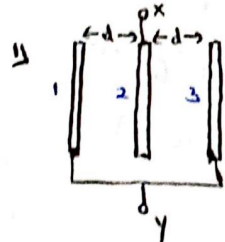
4) If n -capacitors are connected in parallel, then

$$C_{eq} = nC$$

$$V_{each} = V$$

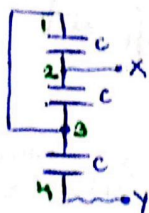
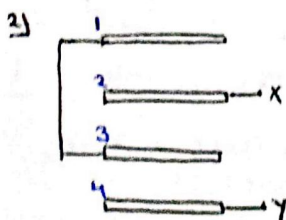
Combination of Parallel plates :-

→ Divide the plate to which terminal is connected into two.



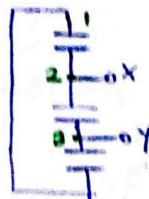
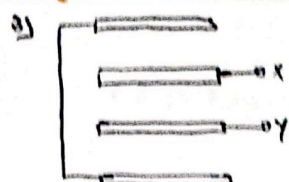
$$C_{eq} = 2C$$

(Parallel connection)



$$C_{eq} = \frac{2C}{3}$$

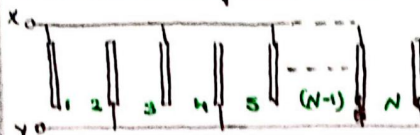
$$C_{eq} = \frac{(2C)(C)}{2C+C}$$



$$C_{eq} = C + \frac{C}{2} \Rightarrow \frac{3C}{2} \quad \underline{\text{Ans}}$$

NOTE: If the plates, use of n plates have connection with n plates $\rightarrow (n-1)$ capacitors.

4) Combination of N -plates

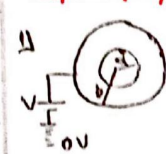


→ All even plates connected together

→ All odd plates connected together

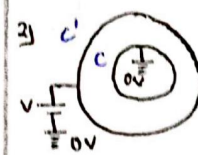
$$C_{eq} = (N-1) C$$

Types of Spherical Shell Capacitor :-



→ Here, only outer shell has potential. So

$$C' = 4\pi \epsilon_0 b$$

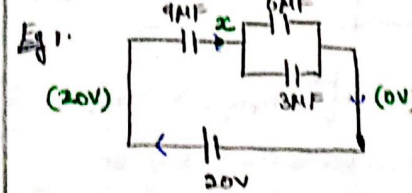


→ Here, C & C' are in parallel connection

$$C_{eq} = C + C'$$

$$C_{eq} = \frac{4\pi \epsilon_0 ab}{(b-a)} + 4\pi \epsilon_0 b$$

Capacitive Circuits :-



$$\text{Sol}^n \text{ (1)} \quad \frac{1}{C_{eq}} = \frac{1}{9} + \frac{1}{(6 \parallel 3)} \Rightarrow \frac{2}{9}$$

$$\Rightarrow C_{eq} = 4.5 \mu F$$

(11) Since both connections of capacitor have same capacitance ($9 \mu F$) in

series combination, voltage is equal

$$V_1 = 10V$$

$$V_2 = V_3 = 10V$$

$$q_1 = C_1 V_1$$

$$q_2 = C_2 V_2 = 60 \mu C$$

$$\Rightarrow q_1 = 90 \mu C$$

$$q_3 = C_3 V_3 = 30 \mu C$$

Nodal Analysis :-

→ Method of potential difference, where we assign potential to every node / terminal.

→ In case of capacitors, we write sum of charge (by $q = CV$) on all plates connected to unknown voltage point equal to zero.

→ (Due to conservation of charge)

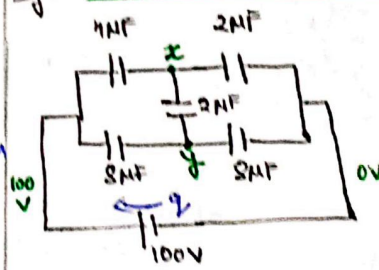
In above example $\Rightarrow$

$$9(x-20) + 6x + 3x = 0$$

$$18x = 180$$

$$\Rightarrow x = 10V$$

Ex2. Unbalanced Wheatstone Bridge.



By Nodal Analysis

$$4(x-100) + 2x + 2y(x-y) = 0$$

$$5x - 2y = 400$$

$$\Rightarrow 4x - y = 200 \quad \text{--- (1)}$$

And,

$$2(y-x) + 8y + 8(y-100) = 0$$

$$18y - 2x = 800$$

$$\Rightarrow 9y - x = 400 \quad \text{--- (2)}$$

From (1) & (2)

$$x = \frac{400}{4}$$

$$y = \frac{360}{7}$$

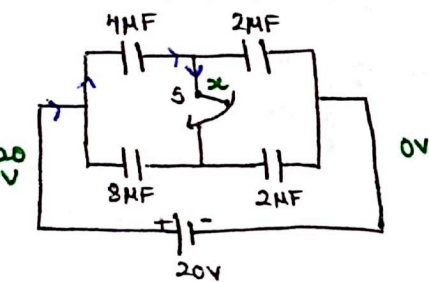
→ We can directly write charge flown by charge conservation on right side capacitors.

$$q_{\text{flown}} = 2x + 8y$$

→ So,

$$C_{eq} = \frac{q_{\text{flown}}}{100}$$

Ex 3: Find charge flown through switch when it is closed.



Solⁿ For point x

$$4(x-20) + 2x + 8(x-20) + 2x = 0$$

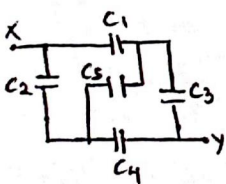
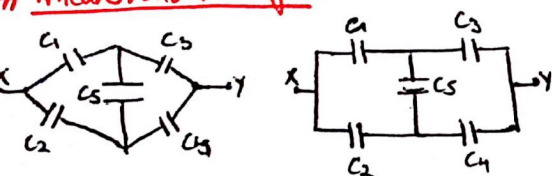
$$\Rightarrow x = 15V$$

So,

$$q_s = 4(x-20) + 2x$$

$$q_s = 4(15-20) + 2(15) = +10\mu C$$

Wheatstone Bridge:



→ All these are different ways to draw wheatstone bridge circuit.

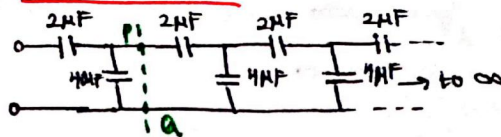
→ For a balanced wheatstone bridge,

$$\frac{C_1}{C_2} = \frac{C_3}{C_4}$$

→ No current will pass through C_5 & the branch can be removed.

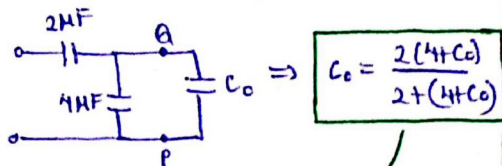
Capacitance - 2

Ladder Networks:



→ Since, ∞ remains ∞ , even after cutting out one segment. So we cut out one segment.

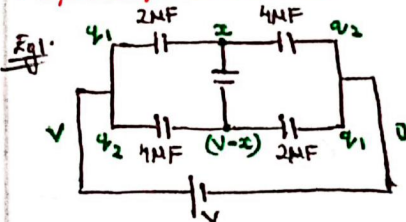
Let, C_{eq} of ∞ segments = C_0



$$C_0 = \frac{2(4+C_0)}{2+(4+C_0)}$$

(Solve this quad. to get value of C_0)

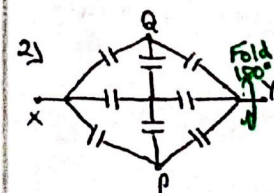
Symmetry in Circuits:



→ Since arrangements of capacitors is symmetric, charges are distributed in same ratio in both branches.

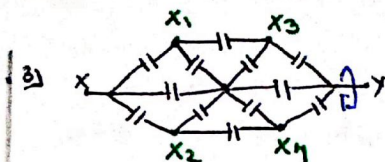
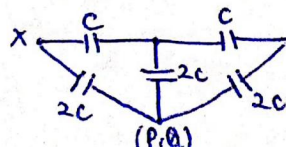
→ So, voltage across (2μF) should also be same in both branches, i.e. $(V-x)$.

→ Hence, only one variable gets introduced.

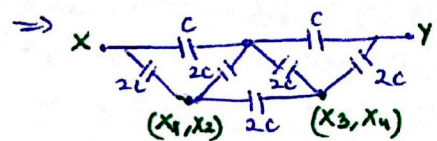


All capacitors have same capacitance (C)

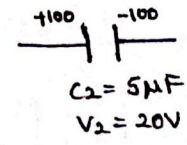
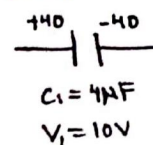
All the overlapping capacitors get connected in parallel.



All capacitors are same.

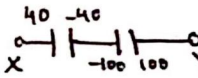


Charge Distribution b/w Capacitors:



1) Series Combination ⇒

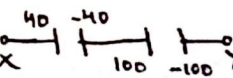
→ Here, charge remain same, no matter how is connection. But eqⁿ of voltage diff. changes.



$$V_x - 10 + 20 = V_y$$

$$V_x - V_y = -10V$$

(Same polarity adjacent)



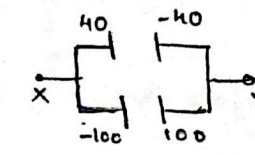
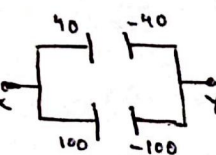
$$V_x - 10 - 20 = V_y$$

$$V_x - V_y = 30V$$

(Opp. Polarity adjacent)

2) Parallel Combination ⇒

→ Here, charge distributed even if battery is not connected.



$$V_x - V_y = \frac{140}{9}$$

(∴ $C_{eq} = 9\mu F$)

$$V_x - V_y = \frac{60}{9}$$

(∴ $C_{eq} = 9\mu F$)

$$\text{Now, } (q_{\text{new}})_{4\mu F} = 4 \times V_{xy}$$

$$(q_{\text{new}})_{5\mu F} = 5 \times V_{xy}$$

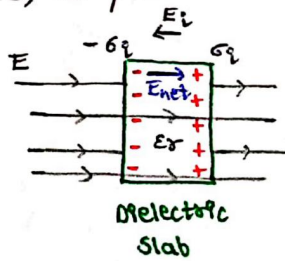
Dielectrics in Capacitors :

→ Dielectrics are insulators whose molecules have tendency to polarise under external EF.

→ The extent of polarisation of dielectric molecules is measured by its 'dielectric constant (ϵ_r/k)'

(For Air/Vacuum, $\epsilon_r = 1$)

(Means, no polarisation takes place)



Dielectric Slab

① Some of the external 'E' gets terminated at -ve surface of dielectric

② But, due to +ve surface of dielectric the final EF again become 'E' on other side.

③ Due to polarisation, charge density arises at boundaries, which is given by

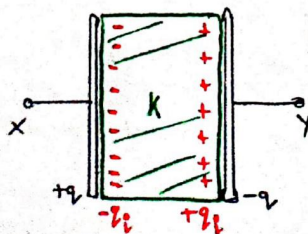
$$\sigma_q = \epsilon_0 E \left(1 - \frac{1}{k}\right)$$

$$E_{net} = \frac{E}{k} = E - \left(\frac{\sigma_q}{\epsilon_0}\right)$$

(Surface Density of bound charges)

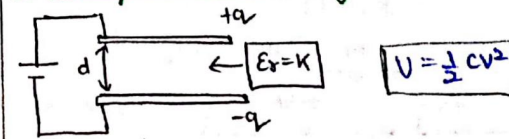
→ When, dielectric is inserted in a charged capacitor, then induced charge of dielectric is given by

$$q_i = q \left(1 - \frac{1}{k}\right)$$



Effect of Dielectric Insertion :

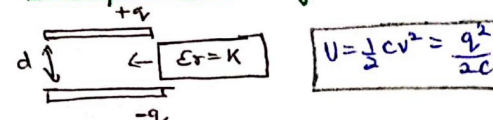
1) Battery connected during Insertion ⇒



Before → q, C, V, E, U

After → $kq, kC, V, E/k, kU$

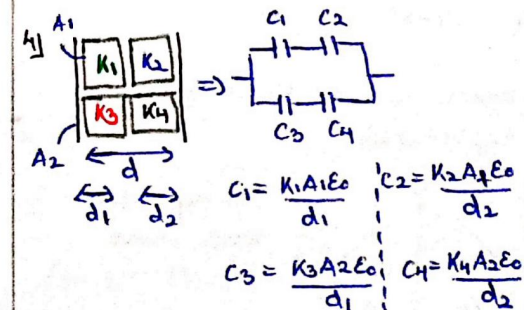
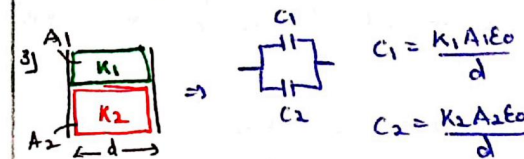
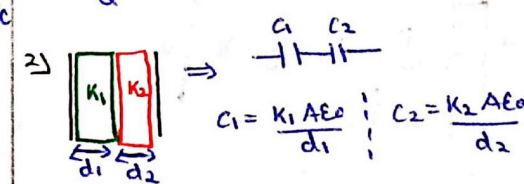
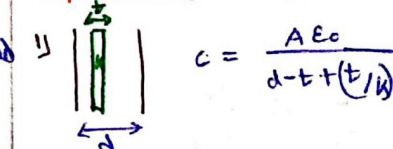
2) Battery absent during Insertion ⇒



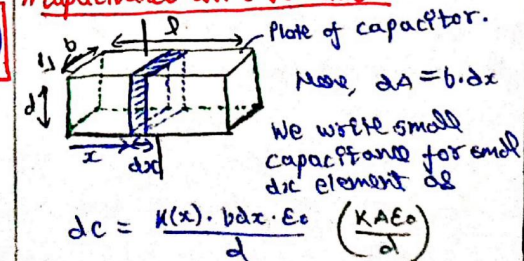
Before → q, C, V, E, U

After → $q, kC, V/k, E/k, U/k$

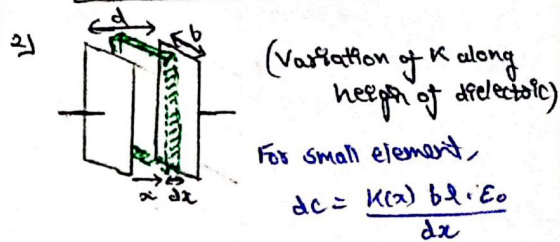
Capacitance for Multiple Dielectrics :



Capacitance with variable k :

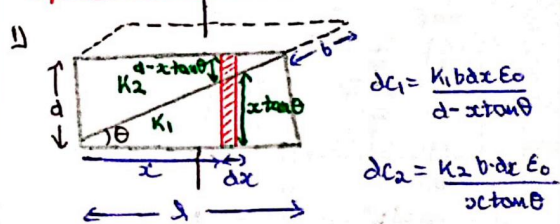


$$C = \frac{\epsilon_0}{d} \int_0^l k(x) \cdot dx \quad (\text{Variation of } k \text{ along length})$$



$$\frac{1}{C} = \int \frac{1}{dC} = \frac{1}{A\epsilon_0} \int_0^d \frac{dx}{k(x)}$$

Capacitance with variable dimensions :



$$So, dC_{eq} = \frac{(dC_1)(dC_2)}{dC_1 + dC_2} \quad C_{eq} = \int_0^l dC_{eq}$$

